TEXORA · LECTURE SHEET

Detailed Study of Fibre Properties

The overview sheet of this course: the full property profile of a fibre — physical, mechanical, moisture, thermal, optical and electrical — set side by side for cotton, wool, polyester, nylon and viscose, and the interrelations that make a fibre's behaviour a single connected story rather than a list.

TE-3209 · TEXTILE PHYSICS · TOPIC 2 · LEVEL 3 TERM 2

1. The property families

A fibre is chosen, processed and judged by a profile, not a single number. The families studied one by one in Topics 3–10 are: **physical/geometric** (length, fineness, density, cross-section, crimp, surface); **mechanical** (tenacity, extension, modulus, elastic recovery, flex and abrasion life); **moisture** (regain, swelling, wet-strength change, heat of sorption); **thermal** (conductivity, softening/melting, heat-setting, flammability behaviour); **optical** (refractive indices, birefringence, lustre); **electrical** (resistance, dielectric behaviour, static proneness); and **chemical/biological** resistance, which processing topics treat in depth elsewhere. This sheet fixes the comparative numbers for the five workhorse fibres; the following topics open each family up.

2. The five fibres side by side

PROPERTY	COTTON	WOOL	POLYESTER	NYLON 6.6	VISCOSE
Density (g/cm³)	1.52	1.30	1.38	1.14	1.52
Tenacity (cN/tex)	27–45	9–13	35–60	40–60	20–25
Breaking extension (%)	5–10	25–35	15–25	15–25	15–25
Commercial regain (%)	8.5	16	0.4	4.5	11
Wet strength vs dry	+10–20%	−15–25%	unchanged	−10%	−40–50%
Birefringence Δn	≈ 0.046	≈ 0.010	≈ 0.16	≈ 0.06	≈ 0.02

Thermal landmark	degrades, no melt	degrades, no melt	melts ≈ 260 °C, Tg ≈ 75 °C	melts ≈ 260 °C, Tg ≈ 50 °C	degrades, no melt
Elastic recovery	low–moderate	excellent	good	excellent	poor

Read the table by columns (a fibre's character) and by rows (a property's league table). Wool is the compliant, thirsty, springy outlier; polyester the strong, dry, heat-settable one; viscose mimics cotton's chemistry without its architecture and pays for it wet; nylon trades a little strength for the best abrasion and recovery in the room.

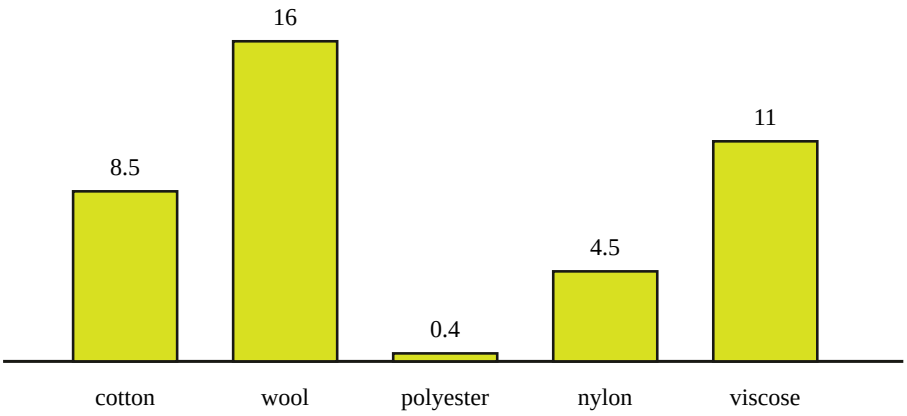


Figure 1 — Commercial (standard) moisture regains of the five fibres, %: wool thirstiest, polyester essentially dry — a single number that predicts dyeing route, static behaviour and comfort.

3. How the properties interrelate

The profile is a web, not a menu. Follow three threads:

- **Moisture is the master variable.** Regain (set by amorphous fraction and polar groups, Topic 1) drives swelling and hence dimensions; plasticises chains and hence modulus, strength and creep; carries ions and hence electrical resistance and static decay (Topics 9–10); releases heat of sorption and hence the warmth of a wool garment put on in the cold; and admits dyes, since dye and water enter the same regions. Contrast a polyester fabric (regain 0.4%): dimensionally indifferent to humidity, static-prone, disperse-dye only, quick-drying — every trait the same dryness in different dress.
- **Orientation and crystallinity trade strength against everything else.** Drawing raises tenacity and birefringence together (Δn is literally the orientation gauge), cuts extension and dye uptake. A high-tenacity yarn and a deep-dyeing yarn are opposing specifications carved from the same polymer.

- **Geometry mediates the rest.** Fineness sets bending stiffness per filament ($\propto d^4$) and therefore fabric softness and cover; density converts between weight-based (tex) and volume-based measures, and decides whether a yarn of given count is bulky (wool, acrylic) or compact (glass, cotton); cross-section shape tunes lustre, soil hiding and cohesion; surface structure — wool's scales, cotton's convolutions — sets friction and felting (Topic 5).

Blending is applied profile arithmetic: polyester–cotton marries polyester's strength, recovery and easy care to cotton's uptake and comfort; wool–nylon rescues wool's abrasion life; elastane at 2–5% rewrites a fabric's recovery column entirely. The blend designer reads the table above like a mixing desk.

4. Key points

- Judge fibres by the whole profile: geometric, mechanical, moisture, thermal, optical, electrical.
- League tables differ per property: wool leads regain and recovery, polyester tenacity-per-price and heat-setting, nylon abrasion, cotton comfort heritage, viscose fluidity and uptake — with a wet-strength warning.
- Regain is the master variable linking moisture, mechanics, dyeing, heat of sorption and static.
- Orientation couples tenacity, birefringence, extension and dye uptake into one trade-off.
- Blending is deliberate profile engineering, not dilution.

5. Practice

Answer in your own words; brief model answers follow.

1. From the table, which fibre would you choose for (a) a swimwear fabric, (b) a winter suit, (c) a workwear overall washed daily — one line of justification each?
2. Convert wool's tenacity of 11 cN/tex to approximate g/den (1 g/den ≈ 8.83 cN/tex) and comment on why wool fabrics still perform respectably.
3. A fabric must dye to a deep shade easily and dry quickly. Explain the conflict in profile terms and name a blend that compromises well.
4. Why does nylon's low density (1.14) matter for a parachute or a climbing rope, beyond its tenacity?
5. Predict two processing consequences of viscose's −40–50% wet-strength figure for its wet processing line.
6. Elastane has unremarkable tenacity but transforms garments at 3% content. Which column of the profile explains this?

Answer key

1. (a) Polyester (or nylon): near-zero strength loss wet, low uptake, quick-drying; cotton/viscose are sponges, wool heavy when wet. (b) Wool: highest regain with heat of sorption, excellent recovery for tailoring, natural crimp traps insulating air. (c) Polyester or polyester-rich blend: strength wet or dry, abrasion life, easy care; pure cotton wears and shrinks, viscose is at its weakest exactly when washed.
2. 11/8.83 ≈ 1.25 g/den — low. Wool compensates with extension (25–35%) and superb recovery, so its work of rupture and wear life are far better than the bare tenacity suggests; fabric engineering (twist, weave, blends) closes the rest.
3. Deep dyeing wants high amorphous accessibility and uptake (hydrophilic fibres); quick drying wants the opposite (low regain). A polyester–viscose or polyester–cotton blend splits the difference: the cellulosic partner dyes and handles moisture, the polyester limits total uptake so drying stays fast.
4. Specific strength: at equal weight a nylon yarn is stronger, and the article's own weight is payload — a rope or canopy of density 1.14 carries roughly 25% more fibre cross-section per kilogram than one of 1.52, compounding the tenacity advantage.
5. Low wet tensions everywhere (draw ratios and machine loads derated for wet viscose), supported fabric transport (rollers, low-drag guides), and careful squeezing rather than hard nip pressure — plus scheduling: the fibre is never weaker than in the wash/dye range, so that is where breaks and distortions will occur if they occur anywhere.
6. Elastic recovery/extension: elastane extends several hundred percent and recovers almost completely, so a few percent by weight dominate the garment's recovery behaviour — profile columns are not weighted by mass fraction alone.
